ELSEVIER

Statistics & Probability Letters 40 (1998) 203–214

STATISTICS & PROBABILITY LETTERS

Formulae and recursions for the joint distributions of success runs of several lengths in a two-state Markov chain

Qing Han*, Sigeo Aki

Department of Information and Mathematical Science, Graduate School of Engineering Science, Osaka University, 1-3 Machikaneyama-cho, Toyonaka, Osaka 560, Japan

Received October 1997; received in revised form February 1998

Abstract

Let $X_1, X_2, \ldots, X_n$ be a time-homogeneous $\{0,1\}$-valued Markov chain. Let $Y=(Y_1,\ldots,Y_r)$ denote the r-dimensional random vector, where Y_i $(i=1,\ldots,r)$ represents the number of success runs of length k_i $(i=1,\ldots,r)$, $k=(k_1,\ldots,k_r)$. In this paper we obtain exact and recurrence formulae for the probability functions and the probability generating functions of Y, based on four different ways of counting numbers of success runs (i.e. overlapping success runs, non-overlapping runs, the runs with a specified length k or more and the runs with just specified length k). Using our methods, we can obtain the probability function and probability generating function of Y, where $Y_1,\ldots,Y_r$ be counted numbers by different ways.

Keywords: Success runs; Two-state Markov chain; Probability function; Probability generating function; Recurrence formulae; Distributions of order k

1. Introduction

In recent years exact discrete distribution theory related to runs statistics has been developed. These are various definitions of runs. The four most important and frequently used success runs are
(1) $M_{n,k}$, the number of overlapping consecutive k successes until the nth trial (Ling, 1988);
(2) $N_{n,k}$, the number of nonoverlapping consecutive k successes until the nth trial (Feller, 1969);
(3) $G_{n,k}$, the number of success runs of size greater than or equal to k until the nth trial;
(4) $E_{n,k}$, the number of success runs of size exactly k until the nth trial (Mood, 1940).

The probability functions (p.f.s) and the probability generating functions (p.g.f.s) of these runs statistics have been studied by many authors in many situations. Hirano (1986) and Philippou and Makri (1986) derived the p.f. of the $N_{n,k}$ in an independent case. Ling (1988, 1989) studied the p.f. related to the $M_{n,k}$ in independent trials. Hirano et al. (1991) obtained the p.g.f.s and the p.f.s of $M_{n,k}$ and $N_{n,k}$ in an independent case. Aki and Hirano (1993) and Hirano and Aki (1993) obtained the p.g.f.s and the p.f.s of $N_{n,k}$, $M_{n,k}$ and $G_{n,k}$ in a two-state Markov chain. Aki (1985) and Aki and Hirano (1988) discussed the p.g.f.s and the p.f.s

* Corresponding author.

0167-7152/98/$ – see front matter
PII S0167-7152(98)00085-6

of some distributions related to runs in a binary sequence of order k. Schwager (1983) discussed the runs in $v \geqslant 2$ possible outcomes at each trial. Fu and Koutras (1994) and Koutras and Alexandrou (1995) studied the distributions of the most common runs statistics based on a finite Markov chain imbedding technique. Fu (1996) and Han and Aki (1998) discussed the joint distributions of runs in a sequence of multi-state trials. Godbole et al. (1997) obtained the joint distributions of success runs (overlapping and nonoverlapping) of several lengths in independent trials.

In this paper we study the distributions of success runs of several lengths k until the nth trial in the following time-homogeneous two-state Markov chain.

Let $X_0, X_1, \ldots$ be a time-homogeneous $\{0,1\}$-valued Markov chain defined by $Pr(X_0=0)=p_0$, $Pr(X_0=1)=p_1=1-p_0$, and $Pr(X_{i+1}=0|X_i=0)=p_{00}$, $Pr(X_{i+1}=1|X_i=0)=p_{01}$, $Pr(X_{i+1}=0|X_i=1)=p_{10}$, $Pr(X_{i+1}=1|X_i=1)=p_{11}$ for $i=0,1,2,\ldots$, where $p_{00}+p_{01}=1$ and $p_{10}+p_{11}=1$.

We discuss the joint distributions of success runs (i.e. $M_{n,k}$, $N_{n,k}$, $G_{n,k}$ and $E_{n,k}$) of lengths $k=(k_1,\ldots,k_r)$.

Let $Y^{\alpha}_{n,k}=(Y^{\alpha}_1, Y^{\alpha}_2, \ldots, Y^{\alpha}_r)$ be the numbers of success runs of lengths $k=(k_1,k_2,\ldots,k_r)$ in $X_1, X_2, \ldots, X_n$, where $\alpha=M,N,G,E$ denote the ways of counting numbers of success runs, i.e.

$$\alpha=\begin{cases} M & \text{overlapping runs,}\\ N & \text{nonoverlapping runs,}\\ G & \text{the runs of size greater than or equal to } k,\\ E & \text{the runs of size exactly } k. \end{cases}$$

In the independent Bernoulli trial case, the distributions of the $Y^M_{n,k}$ and $Y^N_{n,k}$ be called the *overlapping binomial distribution of order k* and the *binomial distribution of order k* respectively, by Godbole et al. (1997). They obtained the exact formulae and recursions of distributions for $Y^M_{n,k}$ and $Y^N_{n,k}$ in an independent case.

In Section 2 we will obtain the recurrence formulae of probability functions of $Y^{\alpha}_{n,k}$, $(\alpha=M,N,G,E)$ in two-state Markov dependent trials. In Section 3 we will obtain the recurrence formulae of probability generating functions of $Y^{\alpha}_{n,k}$, $(\alpha=M,N,G,E)$. These formulae are convenient for evaluating the p.f.s and the p.g.f.s. At the end of this paper, we obtain the exact formulae of the p.f.s and the p.g.f.s, and discuss that these methods can be used to the success runs which can be counted numbers by different ways.

2. The probability functions

Let $B^{\alpha}_i(n,y)$, $(i=0,1$ and $\alpha=M,N,G,E)$ be the conditional probability of y success runs of lengths k in $X_1,\ldots,X_n$, given that $X_0=i$ $(i=0,1)$. Conveniently, we define $B^{\alpha}_i(0,y)=0$ for $\forall y\neq 0$, and $B^{\alpha}_i(n,y)=0$ for $\forall y \ngeqslant 0$ $(n\geqslant 0)$, $(i=0,1$ and $\alpha=M,N,G,E)$.

We denote that

$$\beta(m,\alpha)=\begin{cases} ((m-(k_1-1))^+,\ldots,(m-(k_r-1))^+) & \alpha=M,\\ \left(\left[\dfrac{m}{k_1}\right],\ldots,\left[\dfrac{m}{k_r}\right]\right) & \alpha=N,\\ (I(m\geqslant k_1),\ldots,I(m\geqslant k_r)) & \alpha=G,\\ (I(m=k_1),\ldots,I(m=k_r)) & \alpha=E, \end{cases} \tag{2.1}$$

where $(m-(k_i-1))^+=\max\{0, m-(k_i-1)\}$, $[m/k_i]=\max\{x\colon x\leqslant m/k_i \text{ and } x\in N\}$, $N=\{0,1,\ldots\}$ and

$$I(s)=\begin{cases} 1 & s \text{ is true,}\\ 0 & \text{otherwise.}\end{cases} \quad (i=1,\ldots,r).$$

Considering where the first "0" occurs, we have

Theorem 2.1. *The above conditional probabilities* $B_i^\alpha(n,y)$ $(i=0,1)$, $(\alpha=M,N,G,E)$ *satisfy the following recurrence relation*:

$$B_i^\alpha(n,0)=1 \quad \text{if } 0\leqslant n<\min\{k_j,\ j=1,\dots,r\},\ (i=0,1), \tag{2.2}$$

$$B_0^\alpha(n,y)=p_{00}B_0^\alpha(n-1,y)+\sum_{m=1}^{n-1}p_{01}p_{11}^{m-1}p_{10}B_0^\alpha(n-m-1,y-\beta(m,\alpha))+p_{01}p_{11}^{n-1}B_1^\alpha(0,y-\beta(n,\alpha))$$

$$\text{if } n\geqslant\min\{k_j,\ j=1,\dots,r\} \text{ and } y\geqslant 0, \tag{2.3}$$

$$B_1^\alpha(n,y)=p_{10}B_0^\alpha(n-1,y)+\sum_{m=1}^{n-1}p_{11}^{m}p_{10}B_0^\alpha(n-m-1,y-\beta(m,\alpha))+p_{11}^{n}B_1^\alpha(0,y-\beta(n,\alpha))$$

$$\text{if } n\geqslant\min\{k_j,\ j=1,\dots,r\} \text{ and } y\geqslant 0. \tag{2.4}$$

Proof. Suppose the first "0" occurs in the $(m+1)$th trial X_{m+1}, $(m=0,1,\dots,n-1)$, then the numbers y of success runs in $X_1,\dots,X_n$ are equal to the numbers of success runs in $X_1,\dots,X_m$ plus the numbers of success runs in $X_{m+2},\dots,X_n$. Because $X_1=\cdots=X_m=1$, the numbers of success runs in $X_1,\dots,X_m$ are $\beta(m,\alpha)$, $(\alpha=M,N,G,E)$. So, the numbers of success runs in $X_{m+2},\dots,X_n$ given that $X_{m+1}=0$, are $y-\beta(m,\alpha)$. Hence, this result is easily checked. □

3. The probability generating functions

Let $\phi_n^\alpha(t_1,\dots,t_r)$ be the probability generating function of the distribution of success runs of lengths k in $X_1,\dots,X_n$, $(\alpha=M,N,G,E)$. Let $\psi_n^{(\alpha,i)}(t)$, $(i=0,1)$ be the probability generating function of the conditional distribution of success runs of lengths k in $X_1,\dots,X_n$, given that $X_0=i$, $(i=0,1)$, $(\alpha=M,N,G,E)$. Let $\phi_{j,n}^{(\alpha,l)}(t)$ be the p.g.f. of the conditional distribution of success runs of lengths k in $X_{j+1},\dots,X_n$, given that a success run of length l is just observed at X_j, $(l=0,1,\dots,j;\ j=1,2,\dots n)$. Obviously, we have

$$\phi_{j,n}^{(\alpha,0)}(t)=\psi_{n-j}^{(\alpha,0)}(t). \tag{3.1}$$

In overlapping runs case (i.e. $\alpha=M$), we have

$$\phi_n^\alpha(t)=p_0\psi_n^{(M,0)}(t)+p_1\psi_n^{(M,1)}(t), \tag{3.2}$$

$$\begin{cases}\psi_n^{(M,0)}(t)=p_{00}\phi_{1,n}^{(M,0)}(t)+p_{01}t_1^{I(1-(k_1-1)>0)}\cdots t_r^{I(1-(k_r-1)>0)}\phi_{1,n}^{(M,1)}(t),\\ \psi_n^{(M,1)}(t)=p_{10}\phi_{1,n}^{(M,0)}(t)+p_{11}t_1^{I(1-(k_1-1)>0)}\cdots t_r^{I(1-(k_r-1)>0)}\phi_{1,n}^{(M,1)}(t),\end{cases} \tag{3.3}$$

$$\begin{cases}\phi_{j,n}^{(M,0)}(t)=p_{00}\phi_{j+1,n}^{(M,0)}(t)+p_{01}t_1^{I(1-(k_1-1)>0)}\cdots t_r^{I(1-(k_r-1)>0)}\phi_{j+1,n}^{(M,1)}(t),\\ \phi_{j,n}^{(M,l)}(t)=p_{10}\phi_{j+1,n}^{(M,0)}(t)+p_{11}t_1^{I((l+1)-(k_1-1)>0)}\cdots t_r^{I((l+1)-(k_r-1)>0)}\phi_{j+1,n}^{(M,l+1)}(t),\\ \qquad (l=1,\dots,j;\ j=1,\dots,n-1),\end{cases} \tag{3.4}$$

and

$$\phi_{n,n}^{(M,l)}(t)=1,\quad (\forall l=0,1,\dots,n). \tag{3.5}$$

Hence, we have

$$\begin{aligned}\psi_n^{(M,0)}(t) &= p_{00}\phi_{1,n}^{(M,0)}(t) + p_{01}t_1^{I(1-(k_1-1)>0)}\cdots t_r^{I(1-(k_r-1)>0)}\phi_{1,n}^{(M,1)}(t)\\ &= p_{00}\phi_{1,n}^{(M,0)}(t) + p_{01}t_1^{I(1-(k_1-1)>0)}\cdots t_r^{I(1-(k_r-1)>0)}\\ &\quad\times(p_{10}\phi_{2,n}^{(M,0)}(t) + p_{11}t_1^{I(2-(k_1-1)>0)}\cdots t_r^{I(2-(k_r-1)>0)}\phi_{2,n}^{(M,2)}(t))\\ &=\cdots\\ &= p_{00}\phi_{1,n}^{(M,0)}(t) + \sum_{m=1}^{n-1} p_{01}p_{11}^{m-1}p_{10}t_1^{(m-(k_1-1))^+}\cdots t_r^{(m-(k_r-1))^+}\phi_{m+1,n}^{(M,0)}(t)\\ &\quad + p_{01}p_{11}^{n-1}t_1^{(n-(k_1-1))^+}\cdots t_r^{(n-(k_r-1))^+}\phi_{n,n}^{(M,n)}(t),\end{aligned}$$

and

$$\psi_n^{(M,1)}(t) = \sum_{m=0}^{n-1} p_{11}^m p_{10}t_1^{(m-(k_1-1))^+}\cdots t_r^{(m-(k_r-1))^+}\phi_{m+1,n}^{(M,0)}(t) + p_{11}^n t_1^{(n-(k_1-1))^+}\cdots t_r^{(m-(k_r-1))^+}\phi_{n,n}^{(M,n)}(t).$$

Similarly, we can easily get $\psi_n^{(\alpha,0)}(t)$, $\psi_n^{(\alpha,1)}(t)$, $(\alpha=N,G,E)$, by changing the form "$t_1^{I((l+1)-(k_1-1)>0)}\cdots t_r^{I((l+1)-(k_r-1)>0)}$" $(l=0,1,\ldots,n-1)$ in (3.3) and (3.4) to

$$\begin{cases} t_1^{I(\frac{l+1}{k_1}\in N)}\cdots t_r^{I(\frac{l+1}{k_r}\in N)} & \alpha=N,\\ t_1^{I(l=k_1-1)}\cdots t_r^{I(l=k_r-1)} & \alpha=G, \quad (l=0,1,\ldots,n-1),\\ t_1^{I_1^*(l)}\cdots t_r^{I_r^*(l)} & \alpha=E,\end{cases} \tag{3.6}$$

where

$$I_j^*(l) = \begin{cases} 1 & l=k_j-1,\\ -1 & l=k_j,\\ 0 & l\neq k_j-1,k_j.\end{cases} \quad (j=1,2,\ldots,r).$$

Denoting $t^y = t_1^{y_1}t_2^{y_2}\cdots t_r^{y_r}$ and noting (3.1) and (3.5), we obtain the following theorem.

Theorem 3.1. *The conditional probability generating functions* $\psi_n^{(\alpha,0)}(t)$ *and* $\psi_n^{(\alpha,1)}(t)$ $(\alpha=M,N,G,E)$ *satisfy the recurrence relation*:

$$\psi_n^{(\alpha,0)}(t) = p_{00}\psi_{n-1}^{(\alpha,0)}(t) + \sum_{m=1}^{n-1} p_{01}p_{11}^{m-1}p_{10}t^{\beta(m,\alpha)}\psi_{n-(m+1)}^{(\alpha,0)}(t) + p_{01}p_{11}^{n-1}t^{\beta(n,\alpha)}, \quad (n\geq 1), \tag{3.7}$$

$$\psi_n^{(\alpha,1)}(t) = \sum_{m=0}^{n-1} p_{11}^m p_{10}t^{\beta(m,\alpha)}\psi_{n-(m+1)}^{(\alpha,0)}(t) + p_{11}^n t^{\beta(n,\alpha)}, \quad (n\geq 1), \tag{3.8}$$

where $\psi_0^{(\alpha,0)}(t)=1$, $\psi_0^{(\alpha,1)}(t)=1$, *and*

$$\phi_n^{(\alpha)}(t) = p_0\psi_n^{(\alpha,0)}(t) + p_1\psi_n^{(\alpha,1)}(t).$$

Proof. We can also get Theorem 3.1, using Theorem 2.1 and

$$\psi_n^{(\alpha,0)}(t) = \sum_{y\geq 0} B_0^{(\alpha)}(n,y)t^y,$$

$$\psi_n^{(\alpha,1)}(t)=\sum_{y\geqslant 0}B_1^{(\alpha)}(n,y)t^y. \qquad \square$$

Remark 3.1. Let $r=1$, Theorem 3.1 are the results of Hirano and Aki (1993) and Aki and Hirano (1993).

Remark 3.2. Let $p_{00}=p_{10}$ and $p_{01}=p_{11}$, we can obtain the probability function and the probability generating function of the distribution of successes runs of lengths k in i.i.d. case.

It was discussed by Godbole et al. (1997).

Remark 3.3. Let $r=1$ and $k_1=1$, we can obtain the p.f. and the p.g.f. of the distribution of the number of successes in two-state Markov chain.

The rest of this section will be devoted to the double variable generating functions. We define

$$\xi^{(\alpha)}(t,z)=\sum_{n=0}^{\infty}\psi_n^{(\alpha,0)}(t)z^n, \tag{3.9}$$

$$\eta^{(\alpha)}(t,z)=\sum_{n=0}^{\infty}\psi_n^{(\alpha,1)}(t)z^n. \tag{3.10}$$

We have

$$\begin{aligned}
\xi^{(\alpha)}(t,z)&=\sum_{n=0}^{\infty}\psi_n^{(\alpha,0)}(t)z^n\\
&=\psi_0^{(\alpha,0)}(t)+\psi_1^{(\alpha,0)}(t)z+\sum_{n=2}^{\infty}\psi_n^{(\alpha,0)}(t)z^n\\
&=1+[p_{00}\psi_0^{(\alpha,0)}(t)+p_{01}t^{\beta(1,\alpha)}]z\\
&\quad+\sum_{n=2}^{\infty}\left[p_{00}\psi_{n-1}^{(\alpha,0)}(t)+\sum_{m=1}^{n-1}p_{01}p_{11}^{m-1}p_{10}t^{\beta(m,\alpha)}\psi_{n-(m+1)}^{(\alpha,0)}+p_{01}p_{11}^{n-1}t^{\beta(n,\alpha)}\right]z^n\\
&=1+p_{00}\psi_0^{(\alpha,0)}(t)z+p_{01}t^{\beta(1,\alpha)}z+\sum_{n=2}^{\infty}(p_{00}\psi_{n-1}^{(\alpha,0)}(t))z^n\\
&\quad+\sum_{n=2}^{\infty}\sum_{m=1}^{n-1}[p_{01}p_{11}^{m-1}p_{10}t^{\beta(m,\alpha)}\psi_{n-(m+1)}^{(\alpha,0)}(t)z^n]+\sum_{n=2}^{\infty}[p_{01}p_{11}^{n-1}t^{\beta(n,\alpha)}z^n]\\
&=1+p_{00}z+p_{01}t^{\beta(1,\alpha)}z+p_{00}z\sum_{n=1}^{\infty}\psi_n^{(\alpha,0)}(t)z^n\\
&\quad+\sum_{n=2}^{\infty}[p_{01}p_{11}^{n-1}t^{\beta(n,\alpha)}z^n]+\sum_{m=1}^{\infty}\sum_{n=m+1}^{\infty}[p_{01}p_{11}^{m-1}p_{10}t^{\beta(m,\alpha)}\psi_{n-(m+1)}^{(\alpha,0)}(t)z^n]\\
&=1+p_{00}z+p_{00}z[\xi^{\alpha}(t,z)-\psi_0^{(\alpha,0)}(t)z^0]+\sum_{n=1}^{\infty}[p_{01}p_{11}^{n-1}t^{\beta(n,\alpha)}z^n]\\
&\quad+\left(\sum_{m=1}^{\infty}p_{01}p_{11}^{m-1}p_{10}t^{\beta(m,\alpha)}z^{m+1}\right)\left(\sum_{n=m+1}^{\infty}\psi_{n-(m+1)}^{(\alpha,0)}(t)z^{n-(m+1)}\right)
\end{aligned}$$

$$= 1 + p_{00}z\xi^\alpha(t,z) + p_{01}z\sum_{n=1}^{\infty}(p_{11}z)^{n-1}t^{\beta(n,\alpha)}$$

$$+p_{01}p_{10}z^2\sum_{m=1}^{\infty}(p_{11}z)^{m-1}t^{\beta(m,\alpha)}\xi^\alpha(t,z).$$

Hence, we obtain the following theorem.

Theorem 3.2. *The generating functions of the conditional p.g.f.* $\xi^\alpha(t,z)$ *and* $\eta^\alpha(t,z)$ $(\alpha = M,N,G,E)$ *can be written as*

$$\xi^\alpha(t,z) = \frac{1 + p_{01}zA^\alpha(t,p_{11}z)}{1 - p_{00}z - p_{01}p_{10}z^2A^\alpha(t,p_{11}z)}, \tag{3.11}$$

$$\eta^\alpha(t,z) = [p_{10}z + p_{10}p_{11}z^2A^\alpha(t,p_{11}z)]\xi^\alpha(t,z) + [1 + p_{11}zA^\alpha(t,p_{11}z)], \tag{3.12}$$

where

$$A^\alpha(t,p_{11}z) = \sum_{i=1}^{\infty}(p_{11}z)^{i-1}t^{\beta(i,\alpha)}.$$

Remark 3.4. If $r = 1$, $\alpha = G$ or $\alpha = M$, Theorem 3.2 reduces Theorem 2 or Theorem 3 of Hirano and Aki (1993).

By using Theorem 3.2, we can show an explicit form of the conditional p.g.f. $\psi_n^{(\alpha,0)}(t)$ and $\psi_n^{(\alpha,1)}(t)$.

Theorem 3.3. *The conditional p.g.f.s* $\psi_n^{(\alpha,0)}(t)$ *and* $\psi_n^{(\alpha,1)}(t)$ $(\alpha = M,N,G,E)$ *are written explicitly as:*

$$\psi_n^{(\alpha,0)}(t) = \sum_{n_1+2n_2+\cdots+n\cdot n_n = n}\binom{n_1+n_2+\cdots+n_n}{n_1,n_2,\ldots,n_n}p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^n n_i}$$
$$\times p_{11}^{\sum_{i=3}^n(i-2)n_i}t^{\sum_{i=2}^n n_i\beta(i-1,\alpha)} + \sum_{m=1}^{n}\sum_{n_1+2n_2+\cdots+n\cdot n_n = n-m}$$
$$\times\binom{n_1+n_2+\cdots+n_n}{n_1,n_2,\ldots,n_n}p_{00}^{n_1}p_{01}(p_{01}p_{10})^{\sum_{i=2}^n n_i}p_{11}^{m-1+\sum_{i=3}^n(i-2)n_i}$$
$$\times t^{\beta(m,\alpha)+\sum_{i=2}^n n_i\beta(i-1,\alpha)}, \tag{3.13}$$

and

$$\psi_n^{(\alpha,1)}(t) = \sum_{m=0}^{n}\sum_{n_1+2n_2+\cdots+n\cdot n_n = n-m}\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^n n_i}$$
$$\times p_{11}^{m+\sum_{i=3}^n(i-2)n_i}t^{\beta(m,\alpha)+\sum_{i=2}^n n_i\beta(i-1,\alpha)} + \sum_{m=0}^{n-1}\sum_{n_1+2n_2+\cdots+n\cdot n_n = n-m-1}$$

$$\times \binom{n_1+n_2+\cdots+n_n}{n_1,n_2,\ldots,n_n}(p_{10}-p_{00})p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n} n_i}$$

$$\times p_{11}^{m+\sum_{i=3}^{n}(i-2)n_i} t^{\beta(m,\alpha)+\sum_{i=2}^{n} n_i\beta(i-1,\alpha)}. \tag{3.14}$$

Proof.

$$\xi^\alpha(t,z)=(1+p_{01}zA^\alpha(t,p_{11}z))\left(\sum_{i=0}^{\infty}[p_{00}z+p_{01}p_{10}z^2A^\alpha(t,p_{11}z)]^i\right)$$

$$=\left(1+p_{01}\sum_{n=1}^{\infty}p_{11}^{n-1}t^{\beta(n,\alpha)}z^n\right)\left(\sum_{i=0}^{\infty}\left[p_{00}z+p_{01}p_{10}\sum_{n=1}^{\infty}p_{11}^{n-1}t^{\beta(n,\alpha)}z^{n+1}\right]^i\right)$$

After expanding $\xi^\alpha(t,z)$ w.r.t. z, we have (3.13) by taking the coefficient of z^n. And

$$\eta^\alpha(t,z)=\frac{[1+p_{11}zA^\alpha(t,p_{11}z)]+(p_{10}-p_{00})z[1+p_{11}zA^\alpha(t,p_{11}z)]}{1-p_{00}z-p_{01}p_{10}z^2A^\alpha(t,p_{11}z)},$$

we can obtain (3.14), by using the same method. □

Remark 3.5. If $r=1$ and $\alpha=N$, Theorem 3.3 reduces Theorem 3 of Aki and Hirano (1993).

From formulae (3.13) and (3.14), we can obtain an explicit form of $Pr(Y_{n,k}^\alpha=y|X_0=i)$, $(i=0,1)$, because it is the coefficient of $t_1^{y_1}t_2^{y_2}\cdots t_r^{y_r}$ in $\psi_n^{(\alpha,i)}(t)$, $(i=0,1)$, $(\alpha=M,N,G,E)$.

Theorem 3.4. *The conditional probability functions $Pr(Y_{n,k}^\alpha=y|X_0=i)$, $(i=0,1)$, $(\alpha=M,N,G,E)$ can be written as*

$$Pr(Y_{n,k}^\alpha=y|X_0=0)=\sum_1\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n} n_i}$$
$$\times p_{11}^{\sum_{i=3}^{n}(i-2)n_i}+\sum_{m=1}^{n}\sum_2\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}$$
$$\times p_{00}^{n_1}p_{01}(p_{01}p_{10})^{\sum_{i=2}^{n} n_i}p_{11}^{m-1+\sum_{i=3}^{n}(i-2)n_i}, \tag{3.15}$$

where $\sum_1$ is over all non-negative integers $\{n_i\}_{i=1}^n$ satisfying the conditions

$$\sum_{i=1}^{n} in_i=n, \tag{3.16}$$

$$\sum_{i=2}^{n} n_i\beta(i-1,\alpha)=y, \tag{3.17}$$

and $\sum_2$ is over all non-negative integers $\{n_i\}_{i=1}^n$ satisfying the conditions

$$\sum_{i=1}^{n} in_i+m=n, \tag{3.18}$$

$$\sum_{i=2}^{n} n_i\beta(i-1,\alpha)+\beta(m,\alpha)=y. \tag{3.19}$$

$$Pr(Y^{\alpha}_{n,k}=y|X_0=1) = \sum_{m=0}^{n}\sum_{3}\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n} p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n} n_i} \times p_{11}^{m+\sum_{i=3}^{n}(i-2)n_i} + \sum_{m=0}^{n-1}\sum_{4}\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}(p_{10}-p_{00}) \times p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n} n_i} p_{11}^{m+\sum_{i=3}^{n}(i-2)n_i}, \tag{3.20}$$

where $\sum_3$ *is over all non-negative integers* $\{n_i\}_{i=1}^{n}$ *satisfying the conditions*

$$\sum_{i=1}^{n} in_i+m=n, \tag{3.21}$$

$$\sum_{i=2}^{n} n_i\beta(i-1,\alpha)+\beta(m,\alpha)=y, \tag{3.22}$$

and $\sum_4$ *is over all non-negative integers* $\{n_i\}_{i=1}^{n}$ *satifying the conditions*

$$\sum_{i=1}^{n} in_i+m+1=n, \tag{3.23}$$

$$\sum_{i=2}^{n} n_i\beta(i-1,\alpha)+\beta(m,\alpha)=y. \tag{3.24}$$

Proof. We give a combinatorial proof in here.

First, we prove (3.15). We note that any elementary event in $\{Y_{n,k}=y\}$ can be described as follows:

Let n_i $(i=1,\ldots,n)$ and m $(m=0,\ldots,n)$ represent, respectively, the number of pattern "$\overbrace{1\cdots 1}^{i}0$", and the number of "1" after the last "0" in the sequence. A typical sequence is

$$X_0=0|\overbrace{1\cdots 10}^{i}|\cdots|1\cdots 10|\overbrace{1\cdots 1}^{m}.$$

The probability of the pattern "$\overbrace{1\cdots 10}^{i}$" is

$$\begin{cases} p_{00} & i=1,\\ p_{01}p_{11}^{i-2}p_{10} & i=2,\ldots,n,\end{cases}$$

and the number of success runs is $\beta(i-1,\alpha)$. The probability of the last pattern "$\overbrace{1\cdots 1}^{m}$" is $p_{01}p_{11}^{m-1}$, $(m=1,\ldots,n)$, and the number of success runs is $\beta(m,\alpha)$. Then it is easy to see that if the last trial is "0" (i.e. $X_n=0$, $m=0$), $\{Y^{\alpha}_{n,k}=y\}$ if and only if (3.16) and (3.17) hold, and if the last trial is "1" (i.e. $X_n=1$, $m\geqslant 1$), $\{Y^{\alpha}_{n,k}=y\}$ if and only if (3.18) and (3.19) hold. Thus, we have (3.15).

Second, we prove (3.20). We fix a sequence of outcomes $x=(x_1,\ldots,x_n)$. The sequence has one of the following two forms:

(A) the last trial is "1": $X_0=1|\overbrace{1\cdots 1}^{m}|\overbrace{01\cdots 1}^{i}|\cdots|01\cdots 1,$

(B) the last trial is "0": $X_0 = 1|\overbrace{1\cdots 1}^{m}|\overbrace{01\cdots 1}^{i}|\cdots|0$,

where $m=0,1,\ldots,n$ and $i=1,\ldots,n$. The probability of the pattern "$\overbrace{01\cdots 1}^{i}$" (given that the preceding trial is "1") is

$$\begin{cases} p_{10}p_{01}p_{11}^{i-2} & i=2,\ldots,n, \\ p_{10} & i=1, \end{cases}$$

and the probability of the pattern "$\overbrace{01\cdots 1}^{i}$" (given that the preceding trial is "0") is

$$\begin{cases} p_{00}p_{01}p_{11}^{i-2} & i=2,\ldots,n, \\ p_{00} & i=1. \end{cases}$$

Let $\mathscr{A}=\{x_k|x_k=0$ and $x_{k-1}=1, x_{k+1}=0, k=1,\ldots,n-1\}$ and $\mathscr{B}=\{x_k|x_k=0$ and $x_{k-1}=0, x_{k+1}=1, k=1,\ldots,n-1\}$. $x_k\in\mathscr{A}$ is equivalent to "x_k" is a pattern "$\overbrace{\ 0\ }^{i=1}$" next to "1", and $x_k\in\mathscr{B}$ is equivalent to "$\overbrace{x_kx_{k+1}\cdots}^{i\geqslant 2}$" is a pattern "$\overbrace{01\cdots 1}^{i\geqslant 2}$" next to "0". We have (1) every pattern "$\overbrace{\ 0\ }^{i=1}$" contributes a factor

$$\begin{cases} p_{10} & \text{the trial is in } \mathscr{A}, \\ p_{00} & \text{the trial is not in } \mathscr{A}, \end{cases}$$

(2) every pattern "$\overbrace{01\cdots 1}^{i\geqslant 2}$" contributes a factor

$$\begin{cases} p_{00}p_{01}p_{11}^{i-2} & \text{the first trial of pattern is in } \mathscr{B}, \\ p_{10}p_{01}p_{11}^{i-2} & \text{the first trial of pattern is not in } \mathscr{B}. \end{cases}$$

In Form (A), suppose $x_k\in\mathscr{A}$, then $x_l\in\mathscr{B}$ in $\{x_{k+1},\ldots,x_{n-1}\}$ must exist. i.e. If a pattern "$\overbrace{\ 0\ }^{i=1}$" occurs next to "1", a pattern "$\overbrace{01\cdots 1}^{i\geqslant 2}$" must occur following the pattern "$\overbrace{\ 0\ }^{i=1}$". In Form (A), we have $\forall x_k\in\mathscr{A}$, let $x_k\to x_{\min\{l;x_l\in\mathscr{B},l=k+1,\ldots,n-1\}}$, and $\forall x_l\in\mathscr{B}$, $x_l\to x_{\max\{k;x_k\in\mathscr{A},k=1,\ldots,l-1\}}$. Thus, let $n(\mathscr{A})$ and $n(\mathscr{B})$ denote, respectively, the number of trial in $\mathscr{A}$ and $\mathscr{B}$. We have

$$n(\mathscr{A})=n(\mathscr{B}) \quad \text{if } x \text{ has the form (A).} \tag{3.25}$$

Similarly, we can get

$$n(\mathscr{A})=n(\mathscr{B})+1 \quad \text{if } x \text{ has the form (B).} \tag{3.26}$$

Let n_i $(i=1,\ldots,n)$ be the number of pattern "$\overbrace{01\cdots 1}^{i}$", and m $(m=0,1,\ldots)$ be the number of "1" before the first "0" in sequence. We have, the probability of the above form (A) is

$$\sum_{m=0}^{n-1}\sum_{n_1+2n_2+\cdots+n\cdot n_n=n-m}\left[\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}-\binom{n_1+n_2+\cdots+n_n-1}{n_1-1,n_2,\ldots,n_n}\right]$$
$$\times p_{11}^{m}p_{00}^{n_1}\prod_{i=2}^{n}(p_{10}p_{01}p_{11}^{i-2})^{n_i}+p_{11}^{n},$$

the number of success runs is $\beta(m,\alpha)+\sum_{i=2}^{n} n_i\beta(i-1,\alpha)$, and the probability of from (B) is

$$\sum_{m=0}^{n-1}\sum_{n_1+2n_2+\cdots+n\cdot n_n=n-m}\binom{n_1+n_2+\cdots+n_n-1}{n_1-1,n_2,\ldots,n_n}p_{11}^{m}p_{00}^{n_1-1}p_{10}\prod_{i=2}^{n}(p_{10}p_{01}p_{11}^{i-2})^{n_i},$$

the number of success runs is $\beta(m,\alpha)+\sum_{i=2}^{n} n_i\beta(i-1,\alpha)$.

Thus, the $Pr(Y_{n,k}^{\alpha}=y|X_0=1)$, over all non-negative integers $\{n_i\}_{i=1}^{n}$ satisfying the condition

$$\beta(m,\alpha)+\sum_{i=2}^{n} n_i\beta(i-1,\alpha)=y,$$

is

$$\sum_{m=0}^{n}\sum_{n_1+\cdots+n\cdot n_n=n-m}\binom{n_1+\cdots+n_n}{n_1,\ldots,n_n}p_{11}^{m}p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n}n_i}p_{11}^{\sum_{i=3}^{n}(i-2)n_i}$$
$$-\sum_{m=0}^{n-1}\sum_{n_1+\cdots+n\cdot n_n=n-m}\binom{n_1+n_2+\cdots+n_n-1}{n_1-1,n_2,\ldots,n_n}p_{11}^{m}p_{00}^{n_1}(p_{01}p_{10})^{\sum_{i=2}^{n}n_i}p_{11}^{\sum_{i=3}^{n}(i-2)n_i}$$
$$+\sum_{m=0}^{n-1}\sum_{n_1+\cdots+n\cdot n_n=n-m}\binom{n_1+\cdots+n_n-1}{n_1-1,\ldots,n_n}p_{11}^{m}p_{00}^{n_1-1}(p_{01}p_{10})^{\sum_{i=2}^{n}n_i}p_{11}^{\sum_{i=3}^{n}(i-2)n_i}p_{10}.$$

Thus, this completes the proof of the theorem. This also gives more proof of Theorem 3.3. □

Remark 3.6. If $p_{00}=p_{10}$, $p_{01}=p_{11}$ (i.e. in i.i.d. case), let $\alpha=M$ or N, Theorem 3.4 is Theorem 2.2 or Theorem 2.1 of Godbole et al. (1997).

In the last section, using these methods, we can also obtain these formulae for the p.f. and the p.g.f. of $(Y_1,Y_2,\ldots,Y_r)$, where $Y_1,\ldots,Y_r$ be counted numbers by different ways.

For example, we discuss the joint distribution of $(Y_1^M,Y_2^N,Y_3^G,Y_4^E)$. Let $\beta(m)=((m-(k_1-1))^+,[m/k_2],I(m\geqslant k_3),I(m=k_4))$, we have the same results of Theorems 2.1, 3.1–3.4.

4. Numerical example

Theorems 3.3 and 3.4 may be rather complex in that the inner sum is subject to the condition, and is also a function of m. However, they are very helpful for explaining combinatorial meaning. For numerical computation, we think Theorem 3.1 is convenient. As a concrete example, we consider the joint distribution of $N_{n,2}$ and $N_{n,3}$, where $N_{n,2}$ ($N_{n,3}$) is the number of non-overlapping runs of length 2 (3) in $X_1,X_2,\ldots,X_n$.

Assume the probabilities are $p_{00}=0.7$, $p_{01}=0.3$, $p_{10}=0.3$ and $p_{11}=0.7$, and $n=6$. Using Theorem 3.1, the conditional probability generating function is $\psi_6^{(N,0)}(t_1,t_2)=0.050421t_1^3t_2^2+0.202713t_1^2t_2+0.022491t_1^2+0.160230t_1t_2+0.221340t_1+0.342805$. Hence Table 1 gives the conditional distribution of $(N_{6,2},\ N_{6,3})$, given $X_0=0$.

Let $t_2=1$, we have $\psi_6^{(N,0)}(t_1,t_2=1)=0.050421t_1^3+0.225204t_1^2+0.381570t_1+0.342805$. The conditional marginal distribution of $N_{6,2}$, given $X_0=0$, is $Pr(N_{6,2}=3)=0.050421$, $Pr(N_{6,2}=2)=0.225204$, $Pr(N_{6,2}=1)=0.381570$ and $Pr(N_{6,2}=0)=0.342805$.

Similarly, we can get the conditional probability generating function, given $X_0=1$, $\psi_6^{(N,1)}(t_1,t_2)=0.117649t_1^3t_2^2+0.239757t_1^2t_2+0.034839t_1^2+0.160230t_1t_2+0.211260t_1+0.236265$.

Table 1
Analytical procedures used by MAP3S, AIRMoN and NADP

	$N_{6,2}=0$	$N_{6,2}=1$	$N_{6,2}=2$	$N_{6,2}=3$	
$N_{6,3}=0$	0.342805	0.221340	0.022491	0	0.586636
$N_{6,3}=1$	0	0.160230	0.202713	0	0.362943
$N_{6,3}=2$	0	0	0	0.050421	0.050421
	0.342805	0.381570	0.225204	0.050421	1

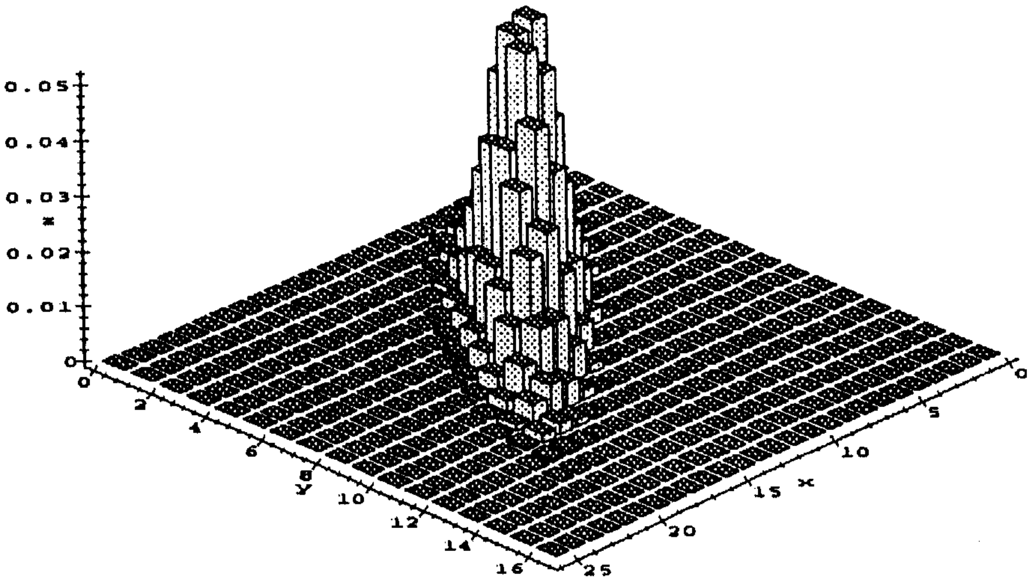

Fig. 1. The conditional exact joint probability function of $(N_{50,2}, N_{50,3})$, given $X_0=0$ and $p_{00}=0.7$, $p_{01}=0.3$, $p_{10}=0.3$, $p_{11}=0.7$.

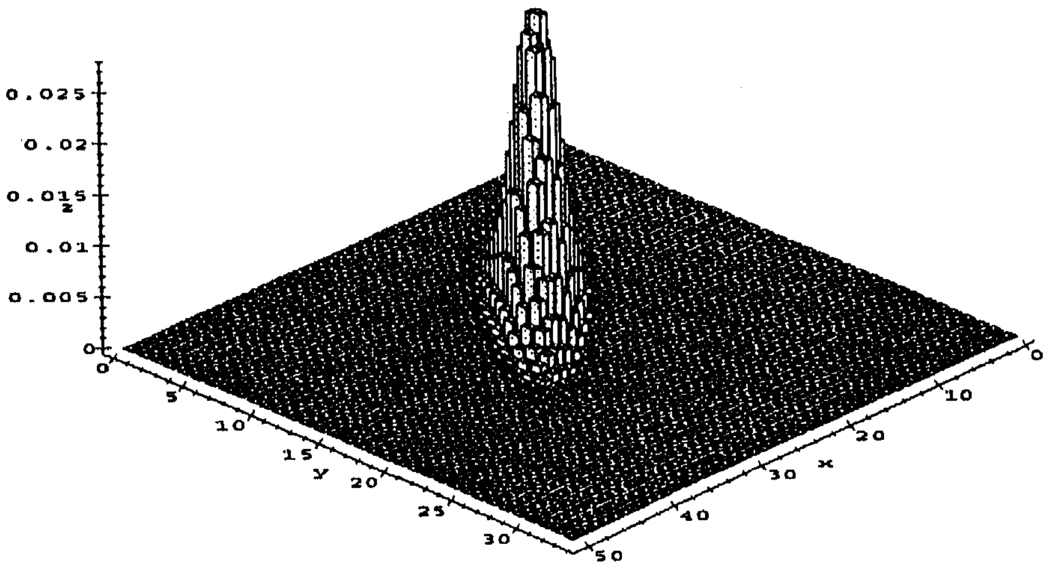

Fig. 2. The conditional exact joint probability function of $(N_{100,2}, N_{100,3})$, given $X_0=0$ and $p_{00}=0.7$, $p_{01}=0.3$, $p_{10}=0.3$, $p_{11}=0.7$.

The proceduce is very simple and suitable for computation by computer algebra systems. For example, for $n=50$ and $n=100$ in the above problem, by using Maple V software, we can get the probability functions. Because the tables are very large, we give Figs. 1 and 2 here, which are the three-dimensional plots of the conditional exact joint probability functions of $(N_{n,2}, N_{n,3})$, given $X_0=0$.

References

Aki, S., 1985. Discrete distributions of order k on a binary sequence. Ann. Inst. Statist. Math. 37, Part A, 205–224.

Aki, S., Hirano, K., 1988. Some characteristics of the binomoal distribution of order k and related distributions. In: Matusita, K. (Eds.), Statistical Theory and Data Analysis II; Proc. 2nd Pacific Area Statistical Conf. Elsevier, Amsterdam, B.V., pp. 211–222.

Aki, S., Hirano, K., 1993. Discrete distributions related to succession events in a two-state Markov chain. In: K. Matusita et al. (Eds.), Statistical Sciences and Data Analysis; Proc. 3rd Pacific Area Statistical Conf. VSP International Science Publishers, Zeist, pp. 467–474.

Feller, W., 1968. An Introduction to Probability Theory and Its Applications, vol. I, 3rd ed. Wiley, New York.

Fu, J.C., 1996. Distribution theory of runs and patterns associated with a sequence of multi-state trials. Statist. Sinica 6, 957–974.

Fu, J.C., Koutras, M.V., 1994. Distribution theory of runs: a Markov chain approach. J. Amer. Statist. Assoc. 89, 1050–1058.

Godbole, A.P., Papastavridis, S.G., Weishaar, R.S., 1997. Formulae and recursions for the joint distribution of success runs of several lengths. Ann. Inst. Statist. Math. 49, 141–153.

Han, Q., Aki, S., 1998. Joint distributions of runs in a sequence of multi-state trials. Ann. Inst. Statist. Math., to appear.

Hirano, K., 1986. Some properties of the distributions of order k. In: A.N. Philloppou, et al. (Eds.), Fibonacci Numbers and their Applications. Reidel, Dordrecht, pp. 43–53.

Hirano, K., Aki, S., Kashiwagi, N., Kuboki, H. 1991. On Ling's binomial and negative binomial distributions of order k. Statist. Probab. Lett. 11, 503–509.

Hirano, K., Aki, S., 1993. On number of occurrences of success runs of specified length in a two-state Markov chain. Statist. Sinica 3, 313–320.

Koutras, M.V., Alexandrou, V.A. 1995. Runs, scans and urn model distributions: a unified Markov chain approach. Ann. Inst. Statist. Math. 47, 743–766.

Ling, K.D., 1988. On binomial distributions of order k. Statist. Pribab. Lett. 6, 247–250.

Ling, K.D., 1989. A new class of negative binomial distributions of order k. Statist. Probab. Lett. 7, 371–376.

Mood, A.M., 1940. The distribution theory of runs. Ann. Math. Statist. 11, 367–392.

Philippou, A.N., Makri, F.S., 1986. Success, runs and longest runs. Statist. Probab. Lett. 4, 101–105.

Schwager, S.J., 1983. Run probabilities in sequences of Markov-dependent trials. J. Amer. Statist. Assoc. 78, 168–175.